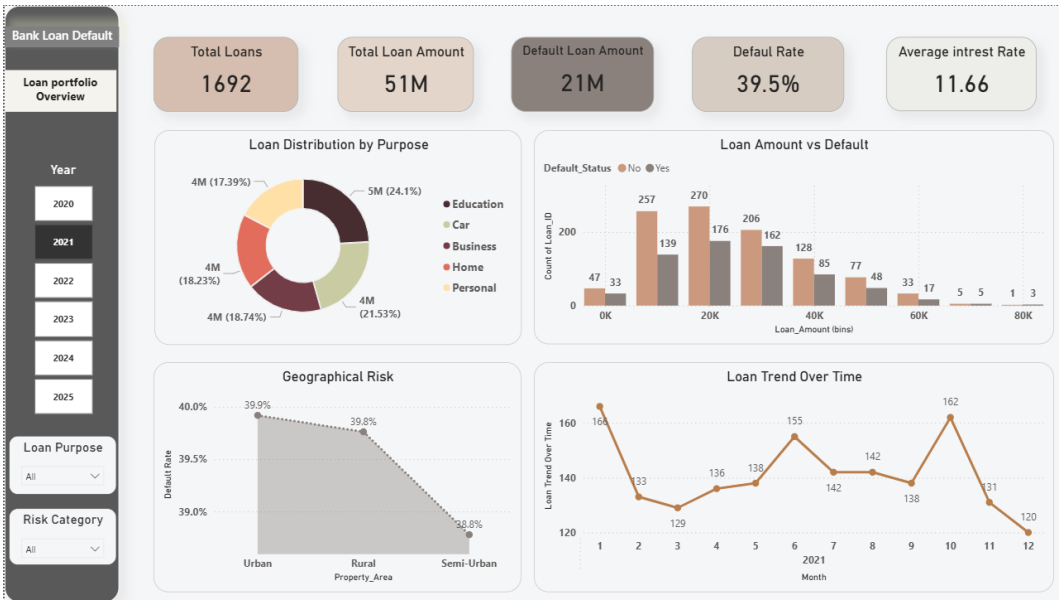
LOAN DEFAULT RISK ANALYSIS DASHBOARD

A Data-Driven Approach to Identify High-Risk Customers

Name: KS VIGNESHWARAN
Institution: IMARTICUS
Course: Postgraduate Program in Data Science and Analytics with GenAI



This project analyzes loan data to identify key factors influencing default risk and helps financial institutions make better decisions.

PROJECT DOCUMENTATION

Project Title

Loan Default Risk Analysis Dashboard

Project Objective

The objective of this project is to analyze loan data and identify key factors influencing loan default risk. The dashboard helps financial institutions understand customer behavior and take proactive measures to reduce defaults.

Dataset Description

The dataset includes customer financial and behavioral information such as:

- Credit Score
- Income
- Loan Amount
- Debt-to-Income Ratio (DTI)
- Late Payments
- Payment Burden Index
- Risk Category
- Default Status

These variables help analyze customer risk and repayment patterns.

Project Steps

1. Data Collection & Preparation

The dataset was collected in CSV format and cleaned by handling missing values, removing duplicates, and standardizing column names.

2. Data Loading & Transformation

The cleaned data was loaded into Power BI. Data types were verified, and transformations were applied using Power Query.

3. Dashboard Development

An interactive dashboard was created using:

- KPI Cards
- Bar Charts
- Line Charts
- Scatter Plots
- Donut Charts

Filters and slicers were added for better user interaction.

4. Insight Generation

The dashboard was used to identify key patterns and relationships affecting loan default risk.

Key Insights

- High DTI customers show higher default rates.
- Late payments are a strong indicator of default risk.
- Low credit score customers are more likely to default.
- Payment burden significantly affects repayment ability.
- High-risk category customers contribute most to defaults.

Dashboard Explanation

The dashboard consists of:

- Portfolio Overview
- Risk Analysis
- Customer Behavior Analysis

Each section provides insights into different aspects of loan performance.

Conclusion

This project demonstrates how data visualization can help identify high-risk customers and improve decision-making. Financial institutions can use these insights to reduce default risk and improve portfolio performance.